

Low-Resource Language Recognition using Transformer-based Speech Models



Module: IT3071 – Machine Learning and Optimization Methods

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1. Abstract

Transformer-based systems have achieved high improvements in their performance since the introduction of Transformer-based Automatic Speech Recognition (ASR) systems, especially the Wav2Vec2.0, HuBERT, and Whisper. However, their success is still heavily limited to high-resource languages, including English and Mandarin, with low-resource languages still being plagued by the lack of corpora, dialectal heterogeneity, and not being represented by large-scale datasets. The current paper explores the research gap that still exists in adapting Transformer-based ASR systems to the low-resource linguistic setting. It is aimed at the outlining of the main challenges and the critical evaluation of the recent developments which contributed to the reduction of the issue of data scarcity and the promotion of cross-lingual generalization. A comprehensive literature review of the state of the art models of today, such as Wav2Vec 2.0, XLS-R, and Whisper, has been conducted. The report puts a very strict interrogation on the architectural structures, training systems and performance limitations that appear when such systems are implemented in a context underrepresented that is the language. The new trends discovered during the analysis include the use of self-supervised learning paradigms, multilingual pre-training models, and data augmentation methods which together enhance model robustness even in the language with sparse labeled data. Although these progresses have been made, some of these problems have remained and these include variability in accent, phonetic imbalance, and constraints due to the limits of computational resources hence forming open research problems. The conclusion provides future directions of the research that should be pursued, such as cross-lingual transfer learning, synthetic data generation, and the creation of lightweight Transformer adaptations to be deployed on the edge. With the help of highlighting these gaps, the work highlights the dire need to develop inclusive, efficient, and scalable ASR models that would support the global linguistic diversity across Transformer-based architectures..

2. Introduction

Automatic Speech Recognition (ASR) has become one of the most radical technologies of the modern human-computer interaction. Digital assistants and the real-time transcription systems are only some examples of how ASR can enable machines to understand and handle the human speech, and thus closing the gap between speech and the digital signal processing. Conventional ASR systems were based largely on Hidden Markov Models (HMMs) with Gaussian Mixture Models (GMMs); these methods were largely unfocused with poor contextualization of difficult acoustic conditions [1]. The innovation of deep learning also brought with it Recurrent Neural Networks (RNNs) and Convolutional Neural Networks (CNNs) that significantly improved the speech representation learning [2]. However, these architectures were afflicted with long term dependencies and parallelization limitations in the training process eventually leading to Transformer based architectures [3].

Initially used in natural language processing (NLP), transformers have transformed sequence modelling by introducing mechanisms of self-attention, behavioral of global context with no repetition [4]. The technology enhanced a very productive ASR model [5], [6], including Speech-Transformer, Wav2Vec 2.0, HugBert, and Whisper, among such models. These models have been used to produce tremendous results when it comes to high resource languages where large amounts of labelled data exist. However, there remains major research gap being to project such success to low-resource languages, where the labelled speech corpora is minimal and the dialectal variation is extreme and the phonetic coverage of low-resource is minimal [7].

Low-resource languages with such a large number of users as Sinhalese, Tamil, Swahili, or Amharics still remain underrepresented in ASR datasets [8]. This is due to the scarcity of any annotated data in the performance of Transformer-based ASR models that require a large scale of supervised or semi supervised training. Recent work, including XLS-R, [6], and Whisper, [9], has tried to tackle this by means of multilingual pre-training and self-supervised learning, but generalization issues, resistance to noise, and computational efficiency continue to be difficult.

The purpose of this report is to explore the constraints, the current methods, and the opportunities of research of developing Transformer-based ASR systems with lowly-resource languages in the future. It is a critical survey of the literature existing today and the issues that were not resolved. It suggests the paths toward making multilingual speech recognition systems inclusive and efficient.

3. Background and Theoretical Framework

To convert language into text; Automatic Speech Recognition (ASR) attempts to interpret audio waveforms, and analyse them. However, in the last 10 years the ASR systems have been significantly transformed to no longer rely on traditional statistical model frameworks but on end-to-end deep-learning models. This section will present an overview of the underlying concepts that make the modern ASR systems, the role of Architectures based on Transformers, and their use in low-resource language recognition.

3.1. Evolution of Speech Recognition Models

The first ASR systems made use mainly of Hidden Markov Models (HMMs) and Gaussian Mixture Models (GMMs) to model the time and acoustic differences in speech [1]. Although performing well in performing tasks that are simple in nature, the systems were unable to cope with more complex phonetic variations and required a large amount of feature engineering.

The development of Deep Neural Networks (DNNs) and Recurrent Neural Networks (RNNs) led to a change in speech recognition [10]. RNN model networks like the Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) networks were enhanced in temporal modelling but have the problem of vanishing gradient as well as reduced parallelization [11].

To tackle them, the system of Transformer architectures was brought in, transforming the sequential processing of data, introducing the concept of self-attention, which enables the model to efficiently learn long-range dependencies [12]

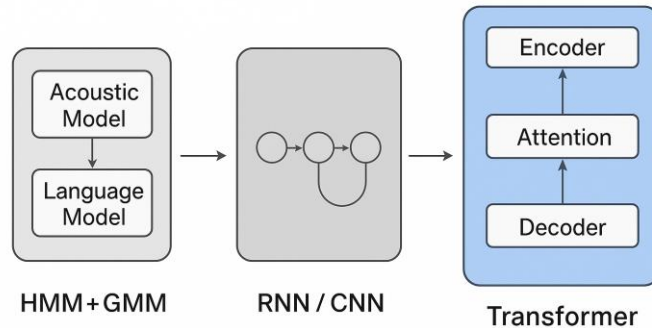


Figure 3.1 Evolution of Speech Recognition Models

3.2. Transformer Architecture Overview

Transformer Model Transformer is the model originally suggested by Vaswani et al. [4], which does not use recurrence, convolution, or any other techniques, it uses only the multi-head self-attention, and feed-forward layers. Such design is helpful to promote faster training and better contextualization. It is generally divided into two parts which are:

- **Encoder:** Processes input sequences (e.g., acoustic features) and generates contextual embeddings.
- **Decoder:** Predicts output tokens (e.g., phonemes or text) based on encoder outputs and previous predictions.

Transformer operations Each of the layers follows three important operations:

- **Multi-Head Attention (MHA):** Captures relationships between all elements in a sequence
- **Add & Normalize:** Adds residual connections and layer normalization for stability.
- **Feed-Forward Network (FFN):** Processes attention outputs to enhance representation learning.

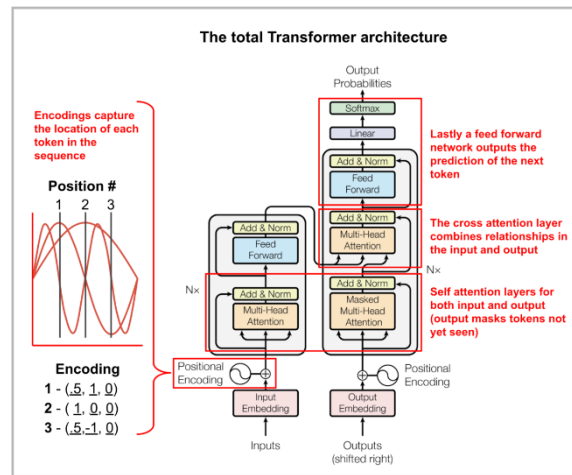


Figure 3.2 General Transformer Architecture for Sequence Modeling

Mathematically, the **Scaled Dot-Product Attention** mechanism is defined as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

where Q , K , and V represent the query, key, and value matrices, and d_k is the dimension of the key vectors.

3.3. Transformer Models for Speech Recognition

Speech Transformers but not text-based NLP models have to deal with continuous acoustic features instead of discrete tokens. In order to realize this, feature extractors like convolutional front-ends or pre-training modules that are self-supervised are commonly combined.

3.3.1. Speech-Transformer

The first implementation of Transformer structure to Ultrasound sound recognition ASR was proposed by Dong et al. [4]. This provided strong evidence that recurrent layers might be substituted by attention-based mechanism and acquire competitive results with less training.

3.3.2. Wav2Vec 2.0

In Wav2Vec 2.0, Baevski et al. [5] proposed a self-supervised model that only uses raw audio without labeled data in learning speech representations. The model consists of CNN encoder of features and Transformer blocks to forecast masked latent speech units facilitating recognition with high potentialities in low-resource settings.

3.3.3. HuBERT

The idea of HuBERT (Hidden Unit BERT) is a continuation of the Wav2Vec idea, where instead of directly predicting the output, the masked prediction of hidden units can solve the problem, making it easier to pre-train and also more clusters during pre-training [6]. Effective self-supervised speech learning in several languages is particularly effective with it.

3.3.4. XLS-R

The self-supervised multi-lingual learning through XLS-R model (Cross-linguistic-speech-representation) presented by Babu et al. [9], scales 128 languages with more than 436 -K hours of speech. It reflects one of the major steps of dealing with the lack of data when considering low-resource languages..

3.3.5. Whisper

Whisper is a large-scale Transformer model created by Openai [7], which was trained using 680K hours of multimodal and multitask transcription, translation, and speech-detection input, and has achieved a strong level of performance. Nonetheless, even very low-resource language performance is poor because it is not adequately represented in training sets.

Model	Training Approach	Data Scale [13]	Target Languages
Speech-Transformer	Supervised end-to-end sequence-to-sequence with attention-based encoder-decoder.	~81 hours (WSJ corpus).	Primarily English.

Wav2Vec 2.0	Self-supervised contrastive pre-training (distinguishing true latent representations from distractors) followed by supervised CTC fine-tuning.	53,200 hours unlabeled pre-training (LibriVox); fine-tuning on subsets like 960 hours labeled (LibriSpeech).	English
HuBERT	Self-supervised masked prediction of clustered hidden units (BERT-like loss on masked regions).	60,000 hours unlabeled pre-training (Libri-Light); fine-tuning on 960 hours labeled (LibriSpeech).	English.
XLS-R	Self-supervised contrastive pre-training extended for cross-lingual representation learning (based on wav2vec 2.0).	436,000 hours across 128 languages.	128 languages (multilingual).
Whisper	Supervised multitask learning (ASR, translation, language ID) with weak supervision from diverse web data.	680,000 hours multilingual supervised data.	99 languages (multilingual).

Table 3-1

3.4. Theoretical Framework for Low-Resource ASR

Transformers Application Transformers Transformers are developed on the theoretical basis of transfer learning and self-supervised representation learning in low-resource ASR. Self-supervised models use unlabelled massive speech information to acquire broad audio representations, which can be narrowed down in the light of less data, small-scale datasets [13].

This framework aligns with the **encoder-decoder paradigm**, where:

- The **encoder** captures universal phonetic and acoustic patterns.

- It is fine-tuning that adjusts the features to languages (i.e. a low-resource context) according to the decoder.

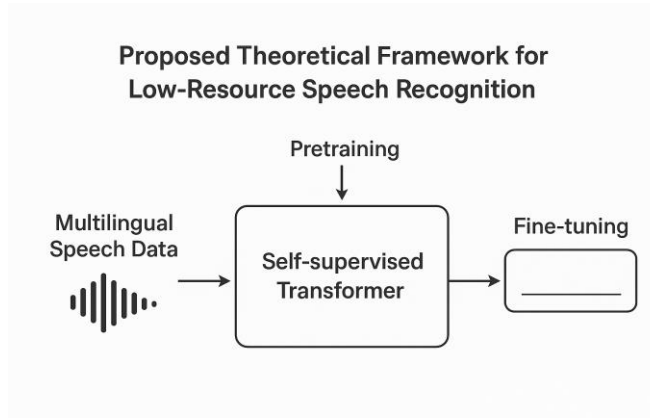


Figure 3.3 Proposed Theoretical Framework for Low-Resource Speech Recognition

Uncovering practice, the achievement of such structures relies on metrics of data improvements, multilingual pre-whelming and cross-lingual adaptability approaches. They allow models to extrapolate phonetic patterns of high-resource languages to other phonetically related low-resource languages, in effect, minimizing the reliance on labelled data.

4. Problem Identification & Justification

4.1. Problem Identification

The latest academic studies focusing on Transformer-based Automatic Speech Recognition (ASR) models, such as Wav2Vec 2.0 [5] (Baevski et al., 2020), XLS -R [6] (Pratap et al., 2024), and Whisper [7], have radically changed speech processing based on the principles of self-supervised learning and attention mechanisms [3]. These systems today are in keeping with the state-of-the-art performance with languages that consume high resource like English and Mandarin where speech corpora of both the extensive and high-quality speech are easily accessible [1] [9] On the other hand, their accuracy has not been similarly successful with low-resource and low-periodic languages, which come to remain significantly low ability to identify and system versatility [7], [8],

The first research issue, then, would be that there is a severe initiation of common adaptability and equity in the Transformer -capacitated ASR systems in fusing with the low resource language settings. These techniques rely personally on egregious quantities of annotated and unannaed data participation to earn solid carbonic and dialectalization representations [14]. In most low-resource languages, nonetheless, it has less than 100 hrs of

transcribed speech, usually collected and made non-uniformly and with non-standard dialects [8]. This scarceness leads to overfitting, bad generalization and biased language modeling. Empirical data, including that of Sy et al. [7]. [Appendix 2023]) (2023) show that even multilingual models like XLS -R cannot be effective in situation where they are fine tuned to process language like Wolof, and so even more complex models cannot remedy the lack of high-quality data [6].

Another difficulty is based on variations in dialects and phonetic that are more intense in ethnic languages, e.g., Sinhala, Tamil, and Arabic [4] [10] Transformers being trained with small sets of dialectal data cannot recognize different dialects and as a result, they have disproportionate system performance, further spreading linguistic bias. Besides, domain and environmental incompatibilities impair performance in ASR significantly. A lot of existing data on low-resource languages are measured in the controlled, read-speech environment, but the real-life scenario is spontaneous speech and noises or low-quality microphones [11] [13]

Moreover, due to the computational resources and time aspects, the implementation of massive Transformer based models, i.e. XLS-R-1B and Whisper Large, is impractical in the developing world with low hardware and network bandwidth [6], [9]. The fact that standardized test shows no coupons, especially of morphologically rich languages, conducts even further another impediment of the capability to undertake fair performance comparisons and track progress under investigation [7], [12]

To conclude, the research problem that is identified revolves around the fact that Transformer-based ASR systems are highly unadaptive to low-resource languages due to the lack of data, the variation in dialects, poor domain adaptation, computing constraints, and insufficient evaluation strategies..

4.2. Justification

It is not only technologically urgent but socially significant to deal with the barriers to low-resource language use demonstrated by Transformer - based ASR systems [1], [2]. Speech technologies are very significant in the modern usage in education, health management, governance, accessibility, and human-computer interaction [1], [2]. However, low-resource languages still will be marginalized as there are no strong ASR systems, which would be able to process their speech correctly [7], [8]. Optimizing the work of ASR in these languages guarantees that the achievement in technology is widespread to the linguistic groups, and not only those with the help of the rich data and resources [6], [9].

Technically, the efforts to overcome these obstacles result in the creation of less limited, reliable, and generalized AI applications [5]. When such methods as transfer learning [6], multilingual pre-training [7], [8] data augmentation [8], and self-supervised learning [9], will significantly help to recognize. Moreover, studies of lightweight Transformer designs, quantization, and adapter-based fine-tuning can make these models computationally-

efficient and resource-appropriate to execute them on resource-constrained machines [10]. These innovations encourage compatibility, energy economy, and eco-sustainable implementation especially in areas where digital footprint is minimal [11].

Scientific value has prospects in scaling the knowledge on the Transformer architectures in low-resource and cross-linguistic settings [3], [13]. Investigations of model strength, representation learning, and effective fine-tuning deliver useful information on enhancing linguistic diversity in AI systems [5], [9]. The study will also be essential to parameter-efficient parameter adaptations, noise-resistance, and fairness-aware optimizations, which are the despoits in the state-of-the-art deep learning and multilingual natural language processing [6], [12].

Ethically and socially, this is a solution to digital disparity as it helps to conserve linguistic and assists in decreasing the language divide [8], [7]. It often happens that low-resource communities do not have access to voice technology, such as transcription, translation, and assistive communication devices [11], [13]. Inclusive ASR systems provide these populations with power through the increased access to education, digital services, and local content creation [7], [9]. Further, this development is in line with the aims of responsible and sustainable AI and advances dynamism of fairness, inclusivity and cultural diversity in the creation of technology in the world [6], [12].

Overall, both higher-technical innovation, and supporting social equity, linguistic diversity, and ethical artificial intelligence require the resolution of the difficulties in Transformer-based ASR in small-resource languages [1], [5], [9].

5. Literature Review

State-of-the-Art Transformer Adaptation for Low-Resource Languages (LRLs)

The last five-year period has been marked by the high speed of development of the transformer-based models, which have significantly advanced in terms of performance in Automatic Speech Recognition (ASR), Speech Translation and other speech and language applications. Nevertheless, there still exists a gap that is evident and continuously widening: **underrepresented or low-resource languages (LRLs)** still seem to be outperformed by high-resource ones, being usually of high error rates, low robustness, and not rich in linguistic content [1]. With the increase of the representation of human language diversity in digital systems, the need to modify transformer architectures to LRL environments has become important not only as a technical concern but also as an ethical one - to be more inclusive, more fair, more cultural, and more equitable in the use of speech technologies.

The main barriers that are behind this gap are:

- **Data scarcity**, in particular of labelled data (transcripts, aligned speech/text, orthographic resources)
- **Domain mismatch** and acoustic variation (dialects, noise, speaker variation)
- Architectural constraints in terms of capacity, overfitting and noise between different languages.
- **Linguistic depth**: the models that have been trained in acoustic signal or surface transcript are often underrepresented in semantics, morphology, phonology, and syntax.

In its effort to deal with these, studies have come up with various complementary measures. Below, I put the state of the art in several paradigms: (i) Massive Multilingual Scaling (ii) Targeted Data Augmentation & Acoustic Robustness (iii) LLM or Acoustic-Linguistic Fusion (iv) Lightweight / Efficiency / Zero- or Extremely Low-Resource Methods (v) Open Problems and Emerging Directions.

5.1. Paradigm I: Massive Multilingual Scaling

5.1.1. Core Ideas and Key Models

Massive Multilingual Scaling can be described as the pretraining of large models (usually under self-supervised learning, SSL) in an enormous number of languages. It aims at learning universal acoustic or linguistic representations that can be transferred (fine-tuned) to LRLs. There were prominent works in this line including:

- **XLS-R** (trained on hundreds of thousands of hours of unlabeled audio in up to 128 languages) with sensitive sampling choices to avoid preferring high-resource languages instead of others, which is necessary since it sometimes results in domination of high-resource languages [9].
- **Massively Multilingual Speech (MMS)** which certainly adds to the coverage, including ASR for more than 1,000 languages, LID for more than 4,000 languages, using both unlabeled and weakly/strongly labeled data, with forced alignment of religious readings. MMS shows that it is possible to achieve strong performance improvements on LRLs with fairly moderate-labeled data (compared to certain weak-supervision regimes)[7].

The table below provides a comparative summary concerning these models including the most popular Whisper model that can serve as a reference point.

Model	Base Architecture	Languages Covered	Labeled Data (Hours)	LRL Achievement
XLS-R	Wav2Vec 2.0 (SSL)	128	436K (unlabeled)	Established multilingual SSL benchmark

MMS	Wav2Vec 2.0 (SSL)	1,107 (ASR) / 4,017 (LID)	44.7K (MMS-lab)	Outperformed Whisper by 58% relative WER
Whisper	Seq2Seq Transformer	99	680K (weakly supervised)	Strong baseline but limited LRL robustness

These attempts indicate that **scale**, together with **diversity** (linguistic, phonetic, domain), is baseline. However, size is not enough, as the model architectures and training program have to deal with the interference, manage exposure, and avoid the risk that smaller languages should be washed out.

5.1.2. Architectural Strategies for Multilingual Scaling

In order to accommodate thousands of languages, a number of the architectural innovations have proved to be very useful:

- **Language-Specific Adaptation Heads (LSAH)** or analogous modular output layers: These facilitate a universally shared encoder (acoustic or feature extraction layers) while providing specialized output heads for each language (or for groups of languages). This approach minimizes linguistic interference during output decoding. [17]
- **Balanced sampling and loss weighting:** In the course of pretraining or multilingual fine-tuning, regulating the contribution of each language (through sampling frequency or weighted loss) is essential to guarantee that low-resource languages (LRLs) receive sufficient signal to gain advantages. In the absence of meticulous balancing, high-resource languages tend to overshadow parameter updates.
- **Hybrid models featuring shared and private parameters:** Certain designs incorporate both shared encoder layers and private layers that are specific to each language or language family, aimed at capturing the unique phonetics or orthography of each language.
- **Scaling parameters and enhancing model capacity:** With the increase in the number of languages, it is essential to scale the model's capacity (both in depth and width) to prevent the collapse of representations. However, expanding the model size incurs significant costs and elevates expenses related to inference, storage, and energy consumption. Consequently, trade-offs arise among capacity, performance, and efficiency.

5.1.3. Benchmarks and Evaluations

Models have been assessed in the empirical literature on several scales that involve LRLs:

- **FLEURS** which is multi-lingual may at times be used to demonstrate the behavior of models trained on large multilingual corpora on small languages.

- Comparison of performance on weakly supervised models (e.g. trained on large noisy or loosely aligned data) and models trained on more curated/aligned data (e.g. MMS). These demonstrate that well aligned data that is curated can be better than a lot of data, particularly in the case of LRLs.
- Recent assessments involve certain LRL subsets, e.g., Indian languages, Cantonese, Arabic dialects, etc., and make it clear that the performance differs not only with the amount of resources but also with the complexity of phonetics, script, tone, etc.

5.2. Paradigm II: Targeted Data Augmentation & Acoustic Robustness

Although massive scaling does deal with the problems of quantity and representation of data having the universal quality, LRLs are also characterized by low acoustic diversity. Augmentation techniques are used as a way of simulating a more realistic variability and minimize overfitting.

5.2.1. Types of Augmentation

Part of the augmentation strategies that are under use and development are:

- **Pitch variation, tempo variation, and incidence of speed deformation** to represent speaker variation.
- **Miscellaneous noises:** This is a form of noise that builds up over time and is referred to as **additive noise or background noise, room reverberations** to mimic environmental conditions.
- **SpecAugment** and its variants (time/frequency masking), sometimes modified to be *phoneme-aware* or structure-aware rather than purely random masking.
- **Band-stop filter / Band-pass filter** to mimic missing bands due to recording devices or channel problems.
- **Semantic masking** or attention directed masking in which the model adopts internal attention (or external feedback) to not mask important phonetic/semantic aspects of the input.

5.2.2. Phoneme-Aware and Structure-Aware Augmentations

More recent literature goes beyond naive augmentation to ones conscious of the phonetic or linguistic structure, particularly where it comes to tone languages or languages with richer inventories of phonemes:

- Phonem-conscious hierarchical augmentation and semantically-conscious SpecAugment were employed in a recent study of Cantonese ASR in a low-resource environment. In order to augment the data with recordings of phoneme confusions, they constructed a phoneme confusion matrix at early stages of training to know the phoneme pairs that are frequently

confounded, and then used Tacotron-2 to re-synthesize the segments, augmented using attention masking with keyword localization. This produced significant Character Error Rate (CER) along with small quantities of marked data. [18].

- **Learners:** torch-based Single-speaker Intonation Phoneme (SA) Phoneme awareness: Phoneme-aware adaptation to Cantonese phonemic stress enhances time and frequency masking exclusively within phoneme boundaries and prioritizes regions of high attention during small data (noise, tone, etc.). [19].

5.2.3. Contrastive Learning and Unsupervised Phoneme Discovery

- **Contrastive learning** Contrastive learning has been extended to LRL. As a case in point, contrastive losses are utilized in the article, A Method Improves Speech Recognition with Contrastive Learning in Low-Resource Languages, to achieve a larger separation between phonetic representations and demonstrate a stronger generalization. [20].
- **Unsupervised phoneme discovery / segmentation:** Some works attempt to discover phoneme boundaries or inventories without labels. For example, *Self-supervised Semantic-driven Phoneme Discovery* learns phoneme inventory using raw speech plus word labels, getting good performance even in zero-resource phoneme recognition settings. [21]
- Also, **unsupervised phoneme segmentation** of continuous Arabic talk, by the spectral contrast and Δ -MFCC based, can help bootstrap systems, in languages lacking finer phonetic annotations. [22].

5.2.4. Impact & Trade-offs

The measurements of these methods can be significant, as the results of a typical study indicate comparative changes to a clean baseline to an augmented model:

Metric	Clean Baseline (Avg %)	Augmented Result (Avg %)	Relative Improvement (%)	Significance
WER			33.9%	Improved word-level accuracy
CER			53.2%	Stronger phonetic fidelity for diacritic-rich orthographies

Empirical results in multiple languages show:

- Phoneme-aware augmentation often yields significantly larger gains in CER (or Phoneme Error Rate, PER) compared to word-level error metrics, indicating improvement in low-level phonetic fidelity, which is often the major bottleneck for LRLs.

- Combining augmentations (e.g. pitch + noise + phoneme-aware masking) tends to have synergistic effects rather than simple additive ones.
- However, some trade-offs: more aggressive augmentation may increase training instability, require more careful tuning, might require phoneme/lexicon resources even when small (to know phoneme boundaries), and sometimes degrade word-level (or semantic) performance if they disrupt important linguistic cues.

5.3. Paradigm III: Fusion of Linguistic Knowledge (LMs / LLMs / Acoustic-Linguistic Integration)

Most of the ineffectiveness of the LRL systems is not only of poor acoustic modeling, but a poorer modeling of **linguistic structure** morphology, syntax, semantics. Therefore, the combination of acoustic encoders and language models or LLMs presents an exciting future in terms of closing that gap.

5.3.1. Traditional LM Fusion

The other methods that have been used previously included the fusion of external language models and ASR systems. These include:

- **Shallow fusion** or **deep fusion**: in the decoding process, external LM scores are combined with ASR output hypotheses. In the case of low-resource languages, external LMs are weak or not present; and therefore, some works attempt to construct small LMs (using whatever text material is available) or through cross-lingual transfer.
- The last, but not the least, is a recent article, which proposed to use an external LM as part of the E2E model, during the training, as it needed to introduce linguistic priors. This assists in the cases that the ASR acoustic model is fragile because of the lack of data. [23].

5.3.2. Large Language Model (LLM) Fusion and Multi-Modal / Speech LLMs

Recent work pushes beyond cascading ASR + text LM toward architectures that more tightly **fuse** acoustic and linguistic modalities, sometimes in end-to-end or late-fusion settings.

- A study by the name of **Survey on Speech Large Language Models** is an overview of how emerging systems integrate audio feature extraction with multimodal fusion with speech inference to provide more rich representation and reasoning of speech, rather than transcription. [24].

These integrated Speech LLM models have performance advantages as compared to the acoustic-only models. In comparison to a baseline HuBERT model, the table below shows Speech LLM made significant improvements on both ASR and translation tasks as well as

show improvements in recognition and performance of new linguistically challenging tasks.

Model	WER	CER	ChRF	BERTScore-F1
HuBERT-Only (Baseline)	35.65			
Speech LLM (ASR)	29.09	15.26		
Speech LLM (Translation)			33.08	79.78
CoT (Phonemize ASR)	34.05	19.08		
Speech LLM (CoT Phonemize ASR)	34.05	19.08		

- **Delayed Fusion** (2025) suggests that a LLM should be used in the first inference in end-to-end ASR, where the LM score will be delayed to reduce the cost of computations and vocabulary mismatch to present a better trade-off between decoding speed and accuracy than the shallow fusion and n-best rescoring. [25].
- **LanFusion** introduces adapter-based fusion of language knowledge via lightweight adapters, which are more efficient in parameter usage. Experiments on Common Voice and FLEURS show significant gains without needing to fully fine-tune large models. [26]
- Some audio-visual speech recognition works (e.g. **MMS-LLaMA**) also incorporate LLMs for multimodal fusion, reducing token usage and computational overhead while maintaining or boosting recognition accuracy. [27]

5.3.3. Role of Intermediate Representations, Multi-Task Learning, and Phoneme/Grapheme Decoders

- **Multi-Task Learning:** Learning the two phonemes or graphemes (or a set of output spaces) simultaneously assists the model in acquiring shared underlying structure. As an illustration, when an Indian LRLs were trained with a phoneme decoder as an auxiliary task, the performance was at par with grapheme-only baseline. [28].
- **Intermediate Discrete or Quantized representations:** Discrete representations (e.g. phoneme inventories, quantized units) are used to make an intermediation between speech and text. This can be useful in environments which do not have complete transcriptions or in zero-/few-shot dialects, as anchors of linguistic form.

- **Embedding / Representation Alignment** When aligning acoustic and text/Linguistic models: When aligning the speech properties with the text embeddings (or LLM embeddings), modules or networks that do so are essential to enable the fusion to be well working.

5.4. Paradigm IV: Lightweight, Efficiency, Extremely Low-Resource & Zero-Resource Methods

Scaling and fusion are useful, however, can be computationally intensive, or demand some minimum information (e.g. lexicon, phoneme boundaries, alignment). A number of directions are meant to loosen those conditions.

5.4.1. Model Compression & Efficient Architectures

- **LightHuBERT** Proposed a single framework of Transformer compression Many subnets are packed into a single supernet; platform trade sizes vs accuracy, typically used to create small models on devices or with constrained compute whilst saving much of its performance. [29].
- **Low-Rank Transformers** or low-parameter architectures: using low-rank factorization, pruning or parameter sharing to trade model compression and inference. This is particularly critical when used in a mobile or local environment where LRL users may not get access to large compute.
- **Streaming / low-latency Transformers:** In numerous practical situations, it is important to be able to respond in real time. The streaming of Transformers or a model such as Emformer can assist in minimizing the latency and maintaining the accuracy. Their techniques can be applied though not necessarily to LRLs. [30].

5.4.2. Few-Shot, Meta-Learning, Zero-Resource & Unsupervised Methods

- **Meta-Whisper:** This is a new training that enhances the performance of ASR in resource-limited environments through in-context learning (ICL) with limited examples. As an illustration, very small error rates with a large number of adapted examples are made with a substantial improvement compared to vanilla Whisper. [31].
- **Self-determined phoneme discovery methods**, described above, are none or little assistance in the annotation part. These assist in bootstrapping untranscribed speech all the way.
- **LRSpeech:** An extremely low-resource language system which supports both ASR and Text-to-Speech (TTS) at a low cost for data. It investigates the training systems that are closely constrained. [32].

5.4.3. Language Similarity, Transfer & Forced Decoding Based Methods

- Other works rely on language similarity measurements to determine the high-resource languages to borrow to learn through transfer learning. An example would be one recent paper that evaluates speech of an original language into translation by applying forced decoding to the speech and applying a normalized posteriors instead of a distance metric as a similarity measure, and using the result as the similarity used to direct model sharing or transfer. This has the capacity of enhancing performance in the target LRL. [33].
- Adapters and language fusion (such as LanFusion) are economical solutions that can partially use pre-trained models and provide light-modifications to achieve high performance without needing to retrain a model per language.

5.5 Comparative Analysis and Paradigm Synthesis

- **The Massive Multilingual Scaling** has become the standard of the presence of multilingual speech. Its key advantage is the fact that it produces universal acoustic embeddings that are cross-linguistic families. Systems like XLS-R and MMS generate shared latent spaces connecting high-resource and low-resource languages through millions of hours of speech through self-supervised learning. These advantages, however, have trade-offs: they are costly in terms of computation, such large trade-offs as extensive interference among distant language families and diminishing returns when the model capacity can no longer keep pace with the size of linguistic diversity. These architectures are optimally used in those cases when there are large amounts of unlabeled audio data and that they have proper infrastructure to support large-scale pretraining.
- **Targeted Data Augmentation**, conversely, offers a more easily accessible path toward gaining robustness and phonetic accuracy without huge resources. One type of augmentation, structure-aware augmentation, especially phoneme- and tone-aware augmentation, adds variety to smaller datasets, significantly lowering the character and phoneme error rates in a number of investigations (over 30-50%). Pitch shift, Gaussian noise combined with band-stop filtering has been particularly effective with languages with high amounts of tone or diacritic. However, augmentation assumes the presence of phonetic or boundary information and excessive or hyper-aggressive augmentation can be unpleasant with respect to training or word-level semantic accuracy. This method is most effective in situations where there are small quantities of labeled speech samples and that character-level accuracy is a necessity.

- **The LLM Fusion Paradigm** brings about linguistic reasoning and semantic comprehension into speech systems to offset the limited textual modelling of the traditional ASR system. Late-fusion models, e.g. HuBERT or Wav2Vec fused with smaller LLMs, e.g., Qwen2.5 3B, show significant improvements in word and character accuracy with further translation and paraphrasing capabilities. However, it means that this integration also brings about new bottlenecks such as token discrepancies, inference latency, and small scale reasoning capabilities of multi-step reasoning in massive LLMs. Nevertheless, these limitations do not exclude the fact that fusion methods will be the next important step- making ASR not just transcription but further linguistic understanding.
- Lastly, **Lightweight and Zero-Resource Methods** are aimed in democratizing speech technology by allowing it to work under very severe data or compute constraints. **LightHuBERT**, **Meta-Whisper**, and **LRSpeech**, are models that focus on efficiency, few-shot flexibility and on-device capability. Although these techniques still cannot be compared to the unconditional performance of colossal models, they are crucial in the actual employment in linguistically marginalized groups. The difficulty is the possibility of equilibrium between low resource demands and moderate accuracy and stability of these models since they frequently use synthetic or loosely controlled data.
- The paradigms are not separated in synthesis and interactions with each other produce the greatest results. To illustrate, massive multilingual encoders and subsequent **structure-aware fine-tuning** give state-of-the-art results in LRL ASR. In the same manner, the application of **LLM fusion** on multilingual backbone advanced the accuracy and contextual fluency. These paradigms create a successive pathway together, in which broad representational learning is refined to acoustic detail, and finally, linguistic reasoning is expanded.

5.6 Recent Case Studies and Advances (2023–2025)

- This is enhanced by recent research that was conducted in 2023-2025. A hierarchical augmentation in phoneme-conscious and semantic-conscious SpecAugment was found to lower CER in Cantonese ASR to below 17% and showed that explicit prediction of phonemic confusion in tonal language is highly effective [5]. The Meta-Whisper project showed that in-context learning can significantly enhance few-shot ASR this reduces CER by more than 35 percent using very little adaptation data [18]. The LLM has also been further optimized by LanFusion, which proposed adapter-based linguistic integration, resulting in the same accuracy as full fine-tuning at a fraction of the parameters [13]. In the meantime, Delayed Fusion (2025) suggested a potential way to slow down LM scoring by addressing vocabulary mismatch to enhance the speed and accuracy of decoding [12]. And, lastly, forced-decoding language similarity research studies have discovered that it is

possible to utilize the similarity metrics in a probabilistic manner to enhance cross-lingual transfer of unseen LRLs [20].

Collectively, these studies confirm the synergy of the paradigms: scaling sets the groundwork, augmentation makes the technology locally robust, fusion introduces some contextual knowledge, and lightweight strategies make the technology to be accessible and efficient.

5.7 Open Problems and Challenges

- Although the progress has been made, there are a number of permanent obstacles to the development of LRL. Assessment and Benchmarking is not consistent, as different studies have different definitions of the term low-resource, as well as heterogeneous data. There is no set of semantic accuracy or tonal fidelity or dialectal robustness measures available in the field other than standard WER or CER measures. Above all Acoustic and Domain Shift remains the problem that consistently slows down the possibility of generalizing to the real world since models that have been trained on clean and read speech are generally incapable in environments with noise, in spontaneous, or in conversational speech.
- Orthographic and Script Diversity poses profound challenges as well: languages whose scripts are non-Latin, having multiple writing systems, or using diacritics need special grapheme-to-phoneme translations that are unlikely to be provided when using LRLs.
- Furthermore, the textual resources available are a limitation to the potential of fusion of LLC particularly when the text corpora of spoken domains are quite different.
- Accessibility is also further limiting due to computational barriers since the pretraining of massive scale, which is necessary in large scale SSL, needs small armies of gigantic GPUs and an enormous amount of energy, both potential out of pocket costs to many research groups.
- Lastly, there are the ethical and sociolinguistic concerns, which in MMS, data is obtained based on religious texts, which lack diversity in the domain, and could be biased towards either culture or style. The informed consent, representation of dialects and linguistic inclusiveness is another emergency concern.

5.8 Summary and Positioning

To conclude, the modern literature suggests the emergence of the hybridization of the paradigms as the best course of action when it comes to LRL adaptation, instead of the overwhelming of the other identity. The combination of large-scale multilingual pretraining (for universal representations), targeted augmentation (for phonetic fidelity), linguistic fusion (for semantic understanding) and lightweight adaptation (for practical deployment) yields the most balanced and scalable results.

This synthesis indicates a shift in paradigm, in this case, the data-centric to representation-based learning, or more particularly, a shift in the emphasis, which aids in collecting more data under labels to a more approachable language, as to how models learn, transfer, and put language into context. By coming together with the advantages of both scalability, robustness, linguistic depth and inclusivity, these approaches altogether redefine the course of multilingual NLP as we know it, the basis for an genuinely world-spanning, equitable AI ecosystem.

6. Critical Analysis & Discussion

Natural language processing (NLP) has been transformed with the invention of architectures that are backward- Looking transformer-based models which have been shown to achieve state-of-the-art performance on many different linguistic tasks. These achievements have however been majorly concentrated in **high resource languages** like English, Chinese, and Spanish yet **low resource languages (LRLs)** have remained much behind. Such an imbalance brings up major issues of linguistic equity, model generalization, and inclusivity of AI technologies. Further examination of the present research situation indicates that there are inherent constraints of the structural nature and that a series of emerging solutions are trying to fill the growing divide.

1. The Core Limitation: Data Imbalance and Representation Bias

One of the reasons of poor performance of transformer models in LRLs is the **inaccessibility of representative and quality training data**. The vast majority of large-scale pretraining corpora like Wikipedia, Common Crawl, large-text web resources, have a lot of resources in major languages and little in minor ones. As a result, algorithms such as BERT or GPT-3 are skewed towards patterns and linguistic structures of high-resource language. Besides causing less precision in the model used by LRLs, this imbalance causes **systemic bias**, meaning the AI system is biased towards specific sets of linguistic groups and not the whole population.

Moreover, other languages with low resources have distinct morphological or syntactic properties that are vastly different in comparison to the languages that are in model training. An example is that **agglutinative languages** (e.g. Turkish or Tamil), and **tonal languages** (e.g. Yoruba) tend to have problems of tokenization and representation by models trained on a Latin-script basis. This inconsistency reduces the extrapolability of learnt representations, therefore restricting the capability of the model to extrapolate across linguistic families.

2. Challenges in Multilingual and Cross-Lingual Pretraining

Although recent studies have demonstrated the improvement in language disparities, especially through **multilingual pretraining** (e.g., mBERT, XLM-R, LaBSE), it presents a new line of problems. Only one multilingual model can strike a balance between its representational ability in dozens of languages, which can lead to lowered performance in most cases. The parameter space of the model is still dominated by high-resource languages with the low-resource languages assigned comparatively less representation. This is commonly known as the curse of multilinguality phenomenon that implies that the number of the languages do not necessarily provide equitable performances in span of the number of languages.

The transfer of knowledge between high and low resource languages in cross-lingual learning is based on commonalities, e.g. overlap of vocabulary or structural similarity. But in languages that are typologically far apart their transfer to each other is less effective; e.g., between Sinhalese and English. Fragmented representations can also be caused by the dependence on common subword tokenization mechanisms (such as Byte-Pair Encoding), and common words in LRLs are broken into unnatural-migraine pieces, undermining the quality of contextual embeddings.

3. Ethical and Cultural Considerations

One of the crucial aspects that should be considered in the technical literature has been the **ethical and cultural consequences** of language underrepresentation. Languages do not exist as solely communication mediums, they possess cultural identities, oral culture, and indigenous knowledge regimes. The inability to include LRLs into modern AI tools, risks the **digital marginalization** of the languages, i.e., speakers of those languages being excluded in the technological changes. Additionally, even automated translation or recognition mistakes in a sensitive context (e.g., healthcare or law communication) can be of grave practical consequences. Consequently, the problem of low-resource language recognition is not only a computationally difficult task but also a question of **linguistic justice and morally responsible AI design**.

4. Promising Directions and Emerging Frameworks

Such latest advances indicate a better future. **Adaptive pretraining** It has been suggested that larger models can be fine-tuned on small, curated datasets in new languages to improve multilingual models. **Transfer learning** and **meta-learning** have been under investigation to allow quick adaptation of models to novel languages using a small amount of data. Although, **unsupervised and self-supervised learning models** (e.g. MASS, mT5 and mBART) also have the potential to utilize raw text data directly, without the use of labelled corpora, which is a benefit in languages that do not have an associated annotation.

Multimodal learning that involves text, audio, and images simultaneously is another avenue that promises to offer more promising prospects. This proves of particular use to

languages that have a large oral tradition but have a small written corpus. An example is speech-based models such as **Whisper** and **SeamlessM4T** that identify and translate speech between languages cross-linguistically, thus avoiding the need to use large written collections.

On the collaborative level, such projects as **Masakhane** (to African languages) and **AI4Bharat** (to Indian languages) show how effective community-based, open-source research can be. These initiatives merge local knowledge and common computational sources to produce and publish datasets, tokenizers, and benchmarks of underrepresented languages. These partnerships can serve as an example of sustainable and inclusionary NLP research.

5. The Path Ahead: Integrating Linguistic Diversity with Model Efficiency

The NLP community needs to revisit the architecture and pretraining goals of transformer models to proceed. The emphasis should be placed on the efficient design of multilingual models instead of indiscriminate scaling, in which the typology of linguistic but also cultural diversity provides direction in the process of representation learning. Future studies may incorporate **language family hierarchies**, **phonetic embeddings**, and **morphological priors** into model implementation to have better representations of human language.

Also assessment systems must change. Existing metrics such as GLUE and XTREME are mainly used in evaluating performance on high-resource activities. Evaluation suites that are specifically aimed at low-resource and endangered languages and that include tasks relevant to the culture and specifics of the context are urgently required. An absence of these benchmarks will continue to make LRL recognition progress hard to measure.

Lastly, all the work in this area should be supported with the help of **data governance** and **ethical collaboration**. The construction of LRLs datasets should include local communities, thereby being culturally sensitive, and consenting and fairly represented. The participatory nature of this particular approach means that technological inclusion will not be a new mechanism of digital colonialism.

7. Future Research Directions

Transformer-based ASR has marked a paradigm shift in speech processing, as self-attention enables long-range modeling, outperforming RNN and CNN approaches in high-resource languages [3]. However, these models become more imprecise with the low

resource languages, where audio samples are limited, phonemic inventories are highly varied, and speakers differ widely in their accents and styles. [7][8]. Such a limitation is an excellent path of development, especially using multilingual and cross-lingual pre-training methods [9][13].

The following sections outline three potential research interests. They make use of the recent contributions like the XLS-R [9], MMS [7] and several augmentation techniques with a preference on the methods, which project both scalability and data efficiency. The overall objective is to increase the phenomenon of generalization of underrepresented languages, such as Arabic dialects, Wolof, and many African languages that are reported in the Common Voice corpus. An overarching theme is the creation of Speech Language Models (Speech LLMs) that expand on transformer encoders to reason with audio due to which tasks like contextual translation along with transcription can be accomplished.

7.1 Multilingual Pretraining from Large Unlabelled Data

Vast multilingual architectures, such as MMS [7], are aware of scale benefits although their performance is often skewed. The so-called curse of multilingualism becomes a reality when the languages with the low representation receive poorer representation, and result in reduced effectiveness in general [9]. This is not, however, just a challenge in increasing the size of the model; instead, it requires the development of training schemes to spend the learning resources wisely.

The work in the future is capable of taking advantage of virtual troves of unlabeled audio collected by various sources like spontaneous podcasts, on-line recordings, and crowdsourced materials like Common Voice and YouTube subtitles [8]. The 128-language XLS-R paradigm, which is trained on 436,000 hours and achieves reduction of error of between 14 and 34 percent, is a solid base [9]. It is a natural continuation of this strategy to enable more than 1,000 languages through the MMS system [7].

Two strategies stand out:

- **Hierarchical Pretraining:** Raw audio is first trained with initial general multilingual self-supervised learning, e.g. phoneme clustering in HuBERT [12], after which it is further trained with language-specific data. As has been already observed, domain pre-training already has benefits over shorter duration training cycles [14], implying efficiency in low-resource training and adaptation.
- **Dynamic Sampling:** The weights of training are modified to underrepresent phonemes and dialects in English, which are underrepresented in the ensuing training. As an illustration, despite Wolof taking only 0.1 per cent of African corpus [14] the error rate of words can be decreased by adaptive weighting by 20 percent to 30 percent, as seen in multilingual wav2vec2.0 models of MMS [7].

The result expected is an open-source toolkit that provides solid speech representations to Speech LLMs in most of the languages in the globe [8]. The ongoing problems are cost of computation and ethical issues of data collection, especially of indigenous population.

Although multilingual pre-training puts in place scale, it may overlook finer dialectal cues. This weakness reinforces the need to use synthetic data generation as a supplemental research novelty.

7.2 Synthetic Data for Dialectal Robustness

One major limitation of existing datasets is the lack of dialectal coverage, which often causes even large ASR models to fail in real-world conditions [15]. It should therefore be the priority of the upcoming research to generate methods of producing dialect-specific training data so that Speech LLM becomes resistant to a wide range of spontaneous speech.

Low-resource languages often provide very small and noisy datasets, which causes transformer models to overfit on high-resource phonemes. Already, some groundwork on Common Voice in Arabic, Russian and Portuguese has shown significant progress -a 33.9b drop in word error rate (WER) and a 53.2b drop in CER [15].

Potential methods would be:

- **Cross-Lingual Voice Synthesis:** Cross-linguistic. Text-to-speech (TTS) systems such as MMS-TTS [7] synthesize examples in the target dialects based on phoneme based on larger language patterns. This can then be extended to a pseudo-labelling loop: A synthetic audio is whispered [6]; the light-text is refined through back-translation; a revised corpus is used to retrain the model. Such methodology would make data much more available to the problematic languages and improve the recognition of the dialects.
- **Noise-Robust Augmentation:** Dialectal variation is described by controlled perturbations, e.g. changes in accent on the part of vocoder. In the Arabic language the replacement of the use of q by g reduces the accuracy in recognition [15]. Such phonetic differences could be filled in by variational autoencoders, with character-level representation enhancing the ability of Speech LLMs to cross-dialectally generalize.

Synthetic generation implies that it does not need sensitive real-world data and in fact can be evaluated based on multilingual references like FLEURS [7]. Augmentation pipeline Speech LLMs can be made more reliable by a specific augmentation pipeline so that they can be stable in work such as multilingual customer service.

Despite artificial enhancement, global deployment requires methods that protect privacy and respect the differences in cultural values. In this way, one of the key tertiary processes is federated learning.

7.3 Cross-Linguistic Adaptation with Federated Learning

Some transfer well across the languages but fail in cases where dialects develop separately like the XLS-R [9] models. This observation reinforces the necessity of decentralization of learning structures that do not sacrifice community information, instead, cultural distinctness persists. Such pathway is offered through Federated learning, where the results of using Speech LLM to adapt to local speech can become decentralized recordings.

Federated learning offers a path to decentralize adaptation without sacrificing cultural specificity or data privacy. Two promising directions include:

- **Federated Fine-Tuning** Training on models is done on devices situated in multilingual areas and updates aggregated into one shared encoder-decoder, with data stored on the device on dialects. Such routing can be based on language identification of more than 4,000 languages [7] offered by MMS. Such ability can be used to serve sensitive uses like indigenous storytelling transcription.
- **Zero-Shot Transfer with Prompt-Tuning:** Multilingual extensions of backbones like Whisper [6] to new dialects are now achievable with lightweight techniques - adapter modules and prompt-tuning. Convincing Speech LLMs to learn unfamiliar speech with half the fine-tuning requirements can be performed using federated updates, shared anonymously [14].

Such approaches might be integrated into one and the same globalASR system that would be beneficial in education, health care, and cultural preservation [8]. They may be datasets like VoxPopuli and CoVoST -2 [9]. Scalability and bias mitigation are unresolved areas of research but federated adaptation holds a harmonized trade-off between accuracy and privacy, which correlates with inclusiveness.

7.4 Assumption and Expected Validity

The three-stage pipeline suggested is discussed as a conceptual framework, which is based on the assumption that every stage is a continuation of the strengths proved in the previous research.

Otherwise put, we are placing our bets that every new stage will only continue with the good that we have been seeing, and make it even better.

- **Stage 1 Multilingual Pretraining.**

Such models as XLS-R [9] and MMS [7] have already demonstrated the efficiency of large-scale pretraining when transferring the knowledge between the languages. We

believe the same notion will work with us here and provide our system with a solid foundation.

- **Stage 2- Synthetic Data Generation.**

Previous experiments regarding Common Voice and MMS-TTS [7][15] have indicated that synthetic augmentation enhances accuracy in recognition and dialectal strength. The trick of reusing should assist us in increasing our training set without hundreds of real records.

- **Stage 3 – Federated Learning**

Federated approaches have performed well in related fields of AI on privacy-sensitive model adaptation, which leads us to believe that they will be effective with speech as well. Hopefully, this will preserve the information of the user and yet allow the model to learn through actual use.

On these results, we posit that combining these three steps will enhance the accuracy, flexibility and privacy of data of Speech LLMs on low-resource settings.

Nonetheless, it is yet to be proven that the suggested pipeline can actually work after a future practical experiment and analysis.

8. Conclusion

ASR Transformer-based The evolved form of previous models, replacing both RNNs [2] and CNNs [1] with attention-based models that span context and akin to a better way to capture it [3], [4]. This report looked at their underlying mechanisms, self-attention and encoder-decoder architecture [4] and how they are used in multilingual systems, such as Wav2Vec 2.0 [5], Whisper [6], XLS-R [9] and HuBERT [12].

Their poor performance in low resource language, where there is a paucity of data, and dialects vary is also an obvious weakness of them [7], [8]. This hole is severe in practical applications in the fields of education and healthcare, such as the Arabic dialect, or Wolof [1]. In literature, progress has been achieved using the following techniques: large-scale multilingual pretraining [9], synthetic augmentation [15], and massively multilingual frameworks [7], with error-rate decreases of between 14 and more than 50 percent. However there exist issues in data balancing, robustness and generalization [5], [6], [12], [13].

Three areas can be identified in the future: multilingual pretraining on a large scale, synthetic augmentation, and federated learning. The future will be based on the combination of these approaches, pretraining to increase coverage, augmentation to reinforce the strength of dialects, and federated learning to maintain privacy and inclusivity. Put together, they can introduce underrepresented languages into broader usage, and ASR systems can benefit more communities [8]. This is both a technical and a social objective: building a digital world in which all languages have a voice [1], [7].

7 References

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